



# ME6221: Theory of Mechanisms

## Course Project

Submission Date: 28 April, 2024

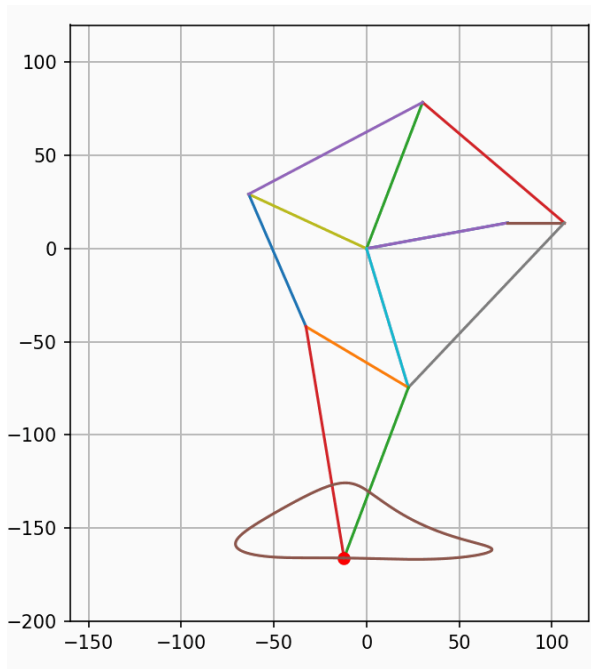
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# Design, Synthesis, and Analysis of The Jansen Linkage



## Problem Statement:

Perform synthesis for 4-position motion generation with prescribed timing for the coupler (foot or the toe) of a Jansen Linkage. To achieve this we have performed the following steps:

- Dimension Synthesis
- Kinematic Analysis
- Dynamic Analysis
- Simulation
- Iteration
- Prototyping

## Application:

Jansen's linkage is a planar leg mechanism used to generate a smooth walking motion. These leg mechanisms have applications in mobile robotics and gait analysis.

## Mathematical Formulation:

It consists of 8 links; 1 ternary and 7 binary (one fixed), hence mobility =  $3(8-1)-2(10) = 1$  (1 DOF mechanism).

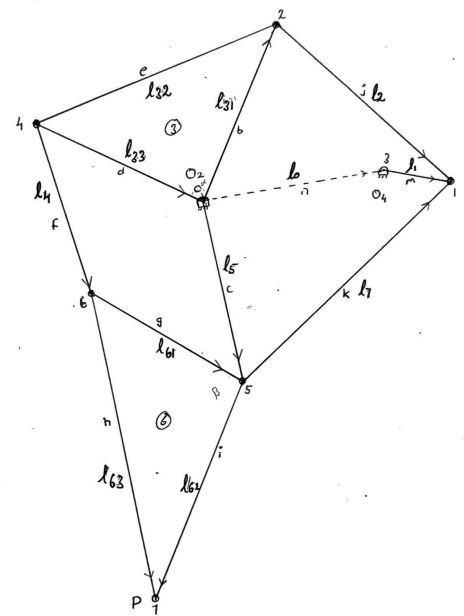
## Dimensional Synthesis:

To synthesize the linkage, we are applying the Burmester theory and Dyad Synthesis formulas learned in class for 4-position motion generation with prescribed timing.

- **Given:**  $\delta_j, \phi_{1j}, \phi_{6j}$  12 scalars

Here  $j = \{2,3,4\}$

The expressions obtained by Dyad Synthesis:



$$z_5(e^{i\phi_{5j}} - 1) + z_{62}(e^{i\phi_{6j}} - 1) = \delta_j \quad (1)$$

$$z_4(e^{i\phi_{4j}} - 1) + z_{63}(e^{i\phi_{6j}} - 1) = \delta_j + z_{33}(e^{i\phi_{3j}} - 1) \quad (2)$$

$$-z_{33} + z_4 + z_{63} = z_5 + z_{62} \quad (3)$$

$$z_{31}(e^{i\phi_{3j}} - 1) + z_2(e^{i\phi_{2j}} - 1) = z_1(e^{i\phi_{1j}} - 1) \quad (4)$$

$$z_5(e^{i\phi_{5j}} - 1) + z_7(e^{i\phi_{7j}} - 1) = z_1(e^{i\phi_{1j}} - 1) \quad (5)$$

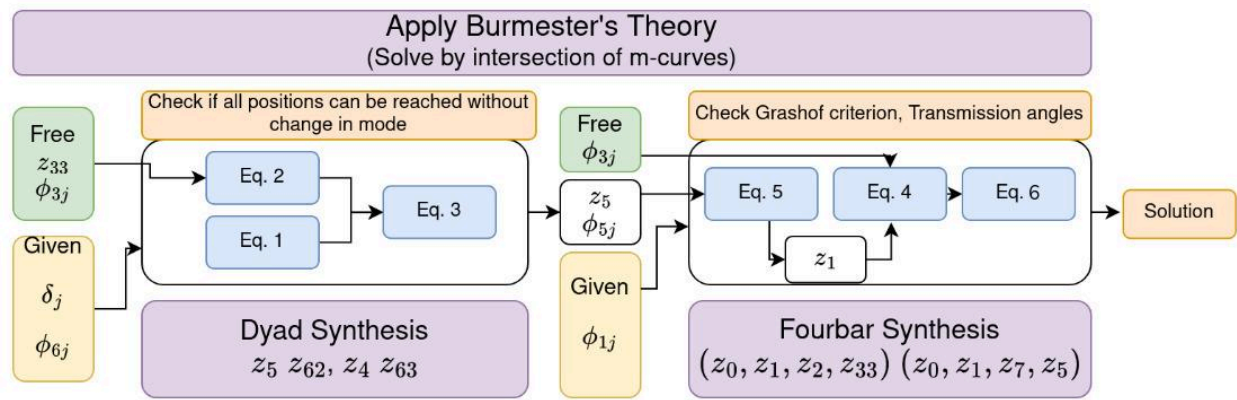
$$z_{31} + z_2 = z_5 + z_7 \quad (6)$$

• **Total Unknowns:** 33 scalars

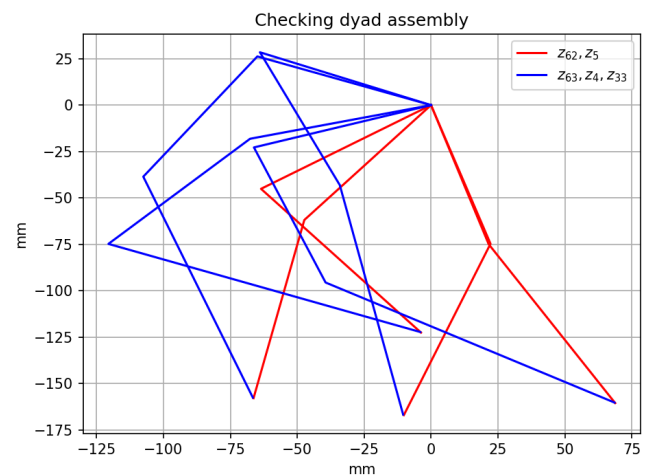
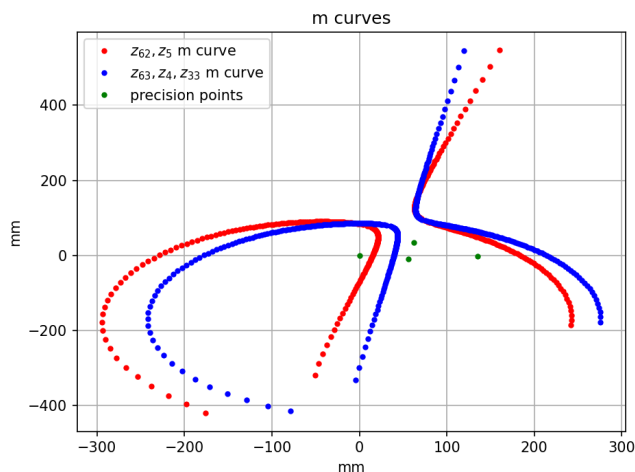
$z_5$     $z_{61}$     $z_7$     $z_{63}$   
 •  $z_1$     $z_{31}$     $z_{33}$     $z_2$     $z_4$   
 $\phi_{5j}$     $\phi_{3j}$     $\phi_{4j}$     $\phi_{2j}$     $\phi_{7j}$

• **Free Choices:**  $33 - 28 = 5$  scalars

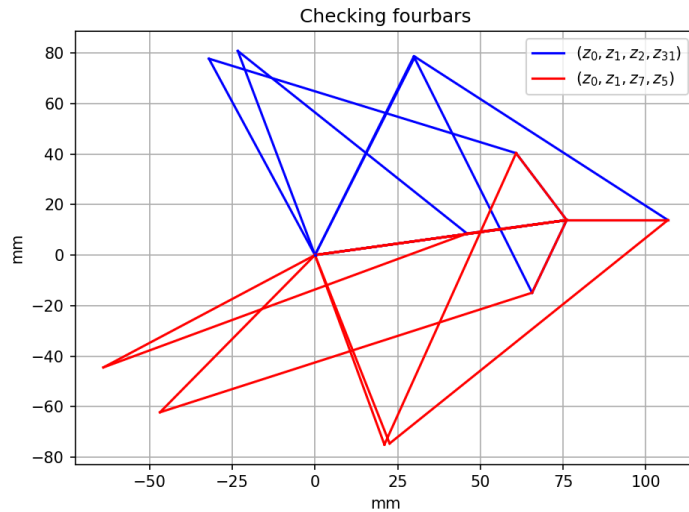
We have taken our free choices as  $z_{33}$   $\phi_{3j}$   $j = \{2,3,4\}$



On solving the following equations, we can plot the Burmester's point pairs for both sets of dyads (Eq. 2, Eq. 1). The curves below show the m-curves that are used to find the center point (the point O2), this satisfies Eq. 3. Similarly the Eq. 5, Eq. 4 are solved to obtain the two 4-bars, and as done earlier the intersection of the m curves for the dyads are found (the point O4) to satisfy Eq. 3.



We have picked our center points such that all positions are reached without mode change (i.e. without branch defect). This can be verified by plotting the dyad assembly at the precision points needed.



This plot of the whole assembly verifies that all precision points can be reached without disassembly in order. The finalized design is:

$$\theta_0 = 0.1799\text{rad}$$

$$\begin{array}{lll} z_{31} = 84.088 & z_5 = 77.909 & z_{33} = 70.000 \\ z_{32} = 105.978 & z_4 = 77.535 & z_{61} = 64.193 \\ z_{63} = 125.849 & z_{62} = 97.838 & z_2 = 100.113 \\ z_7 = 122.038 & z_0 = 77.337 & z_1 = 30.594 \end{array}$$

Grashof criterion is satisfied for both 4-bars, with  $z_1$  as a crank in both cases. This verifies the 4-bar design.

## Kinematic Analysis:

### Position Analysis:

Obtained equations:

$$z_0 + z_1 = z_{31} + z_2 \quad (1) \text{ Vector Loop}$$

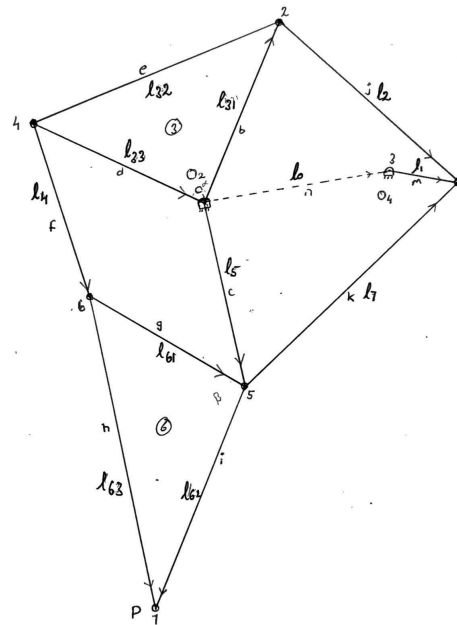
$$z_0 + z_1 = z_5 + z_7 \quad (2) \text{ Vector Loop}$$

$$z_{33} + z_5 = z_4 + z_{61} \quad (3) \text{ Vector Loop}$$

$$\theta_{33} = \theta_{31} + \alpha + \pi \quad (4) \text{ Rigid body}$$

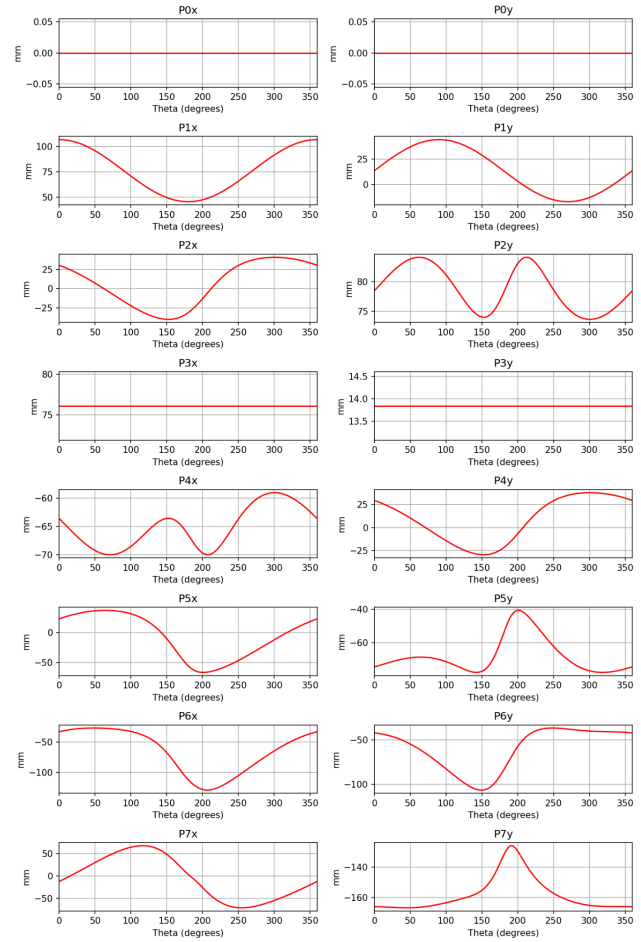
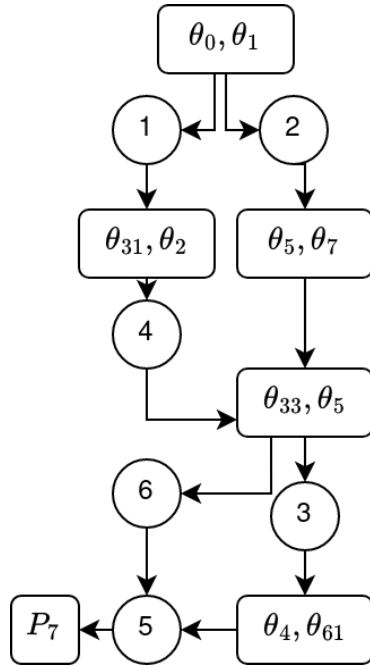
$$\theta_{33} = \theta_{31} + \beta - \pi \quad (5) \text{ Rigid body}$$

$$P_7 = z_5 + z_{62} \quad (6) \text{ Vector Equation}$$



Method of Solving:

This shows the order in which the variables are found.



### Velocity Analysis:

Obtained by differentiating position equations.

Let all the 4 bars be represented by a 4-bar (a,b,c,d).

$$(a, b, c, d) \equiv (z_0, z_1, z_2, z_{31}) \equiv (z_0, z_1, z_7, z_5) \equiv (z_{33}, z_5, z_{61}, z_4)$$

$$ae^{i\theta_a} + be^{i\theta_b} = ce^{i\theta_c} + de^{i\theta_d} \quad \text{Position equation}$$

$$ae^{i\theta_a}\omega_a + be^{i\theta_b}\omega_b = ce^{i\theta_c}\omega_c + de^{i\theta_d}\omega_d \quad \text{Velocity equation}$$

$$\begin{bmatrix} c \cos \theta_c & d \cos \theta_d \\ c \sin \theta_c & d \sin \theta_d \end{bmatrix} \begin{bmatrix} \omega_c \\ \omega_d \end{bmatrix} = \begin{bmatrix} a\omega_a \cos \theta_a + b\omega_b \cos \theta_b \\ a\omega_a \sin \theta_a + b\omega_b \sin \theta_b \end{bmatrix}$$

### Acceleration Analysis:

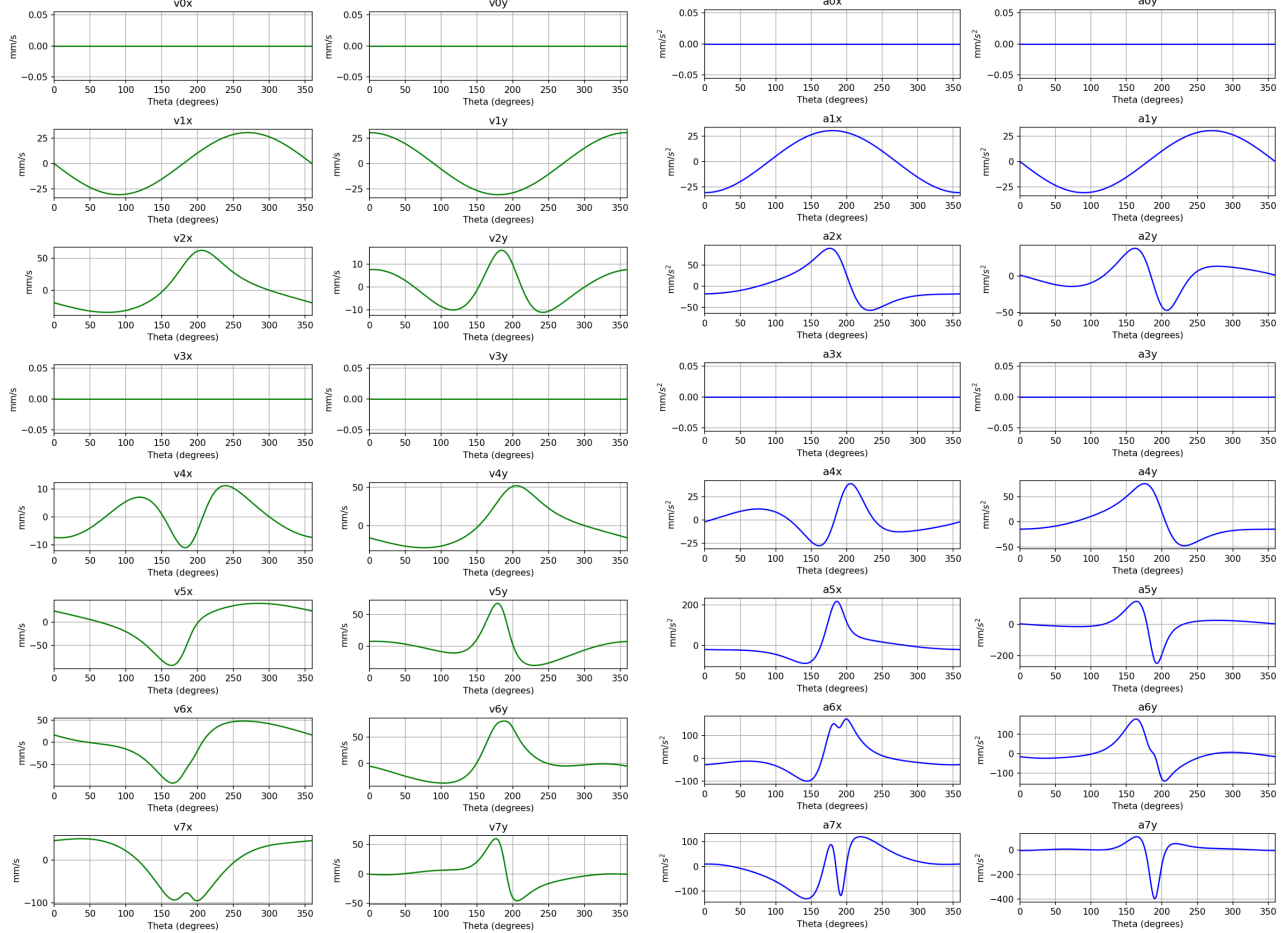
Obtained by differentiating velocity equations.

$$ae^{i\theta_a}(i\omega_a^2 + \alpha_a) + be^{i\theta_b}(i\omega_b^2 + \alpha_b) = ce^{i\theta_c}(i\omega_c^2 + \alpha_c) + de^{i\theta_d}(i\omega_d^2 + \alpha_d) \quad \text{Acceleration equation}$$

$$\begin{bmatrix} c \cos \theta_c & d \cos \theta_d \\ c \sin \theta_c & d \sin \theta_d \end{bmatrix} \begin{bmatrix} \omega_c \\ \omega_d \end{bmatrix} = \begin{bmatrix} a\alpha_a \cos \theta_a + b\alpha_b \cos \theta_b - a\omega_a^2 \sin \theta_a - b\omega_b^2 \sin \theta_b + c\omega_c^2 \sin \theta_c + d\omega_d^2 \sin \theta_d \\ a\alpha_a \sin \theta_a + b\alpha_b \sin \theta_b + a\omega_a^2 \cos \theta_a + b\omega_b^2 \cos \theta_b - c\omega_c^2 \cos \theta_c - d\omega_d^2 \cos \theta_d \end{bmatrix}$$

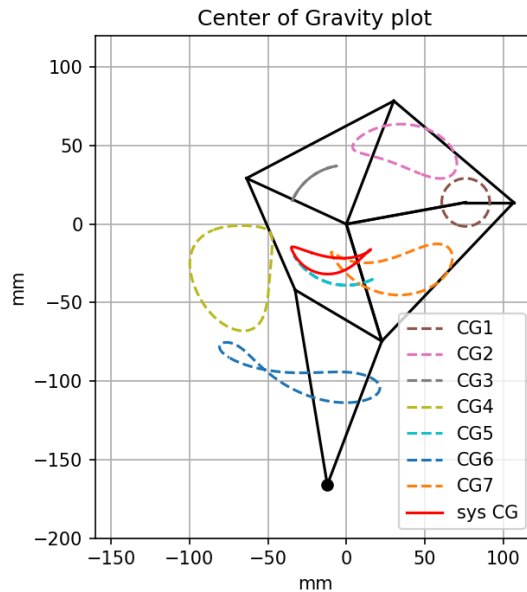
### Joint velocity and acceleration plots:

These are plotted taking a constant input angular velocity = 1 rad/s



### Plot of motion of Center of Gravity of Links and the entire Linkage:

To check stability, we are finding the entire linkage's center of gravity and checking if its range of motion is small.



### Requirements For Synthesis:

1. Grashoff Criterion: Verified for both 4 bar linkages:

For  $z_0z_1z_5z_7$ :  $z_1+z_7 = 30.594+122.038 = 152.632$  mm  
 $z_5+z_0 = 77.909+77.337 = 155.246$  mm

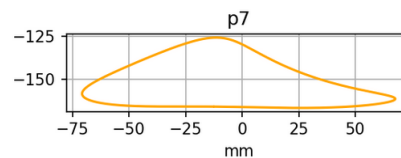
For  $z_0z_1z_2z_{31}$ :  $z_1+z_2 = 100.113+30.594 = 130.707$  mm  
 $z_0+z_{31} = 84.088+77.337 = 161.425$  mm

2. Transmission Angles:

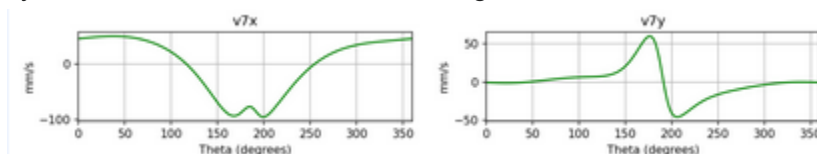
For  $z_0z_1z_5z_7$  (Required > 20): Min: 25° Max: 71°  
 For  $z_0z_1z_2z_{31}$ : Min: 8° Max: 61°

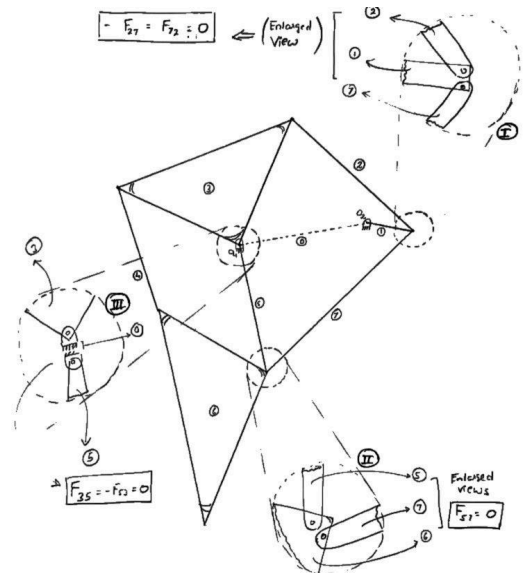
3. Stride Length & Height:

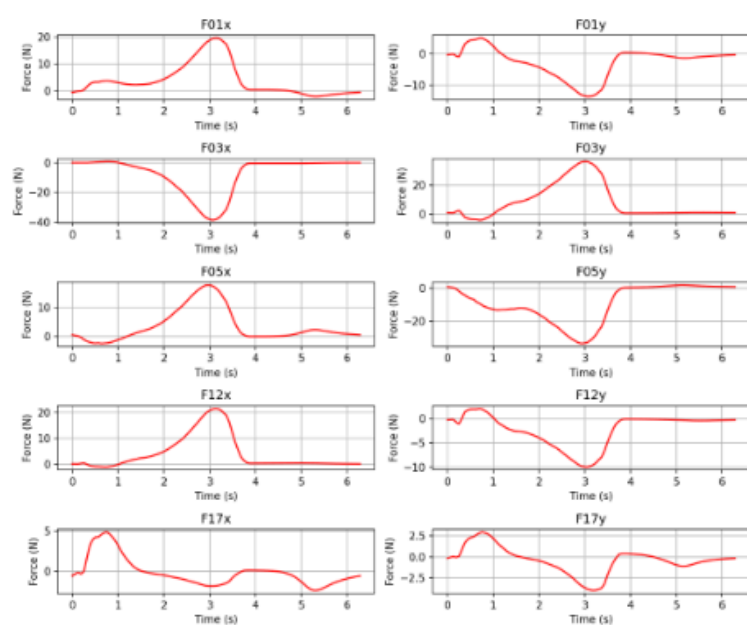
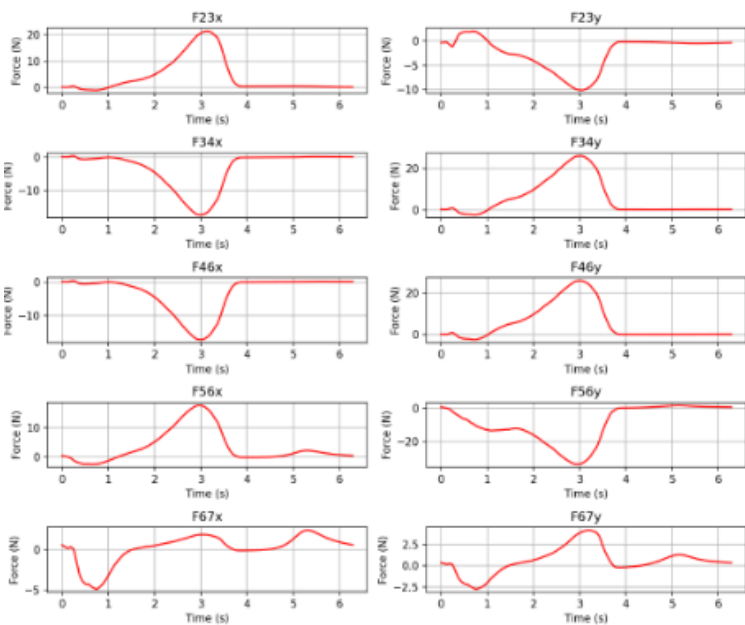
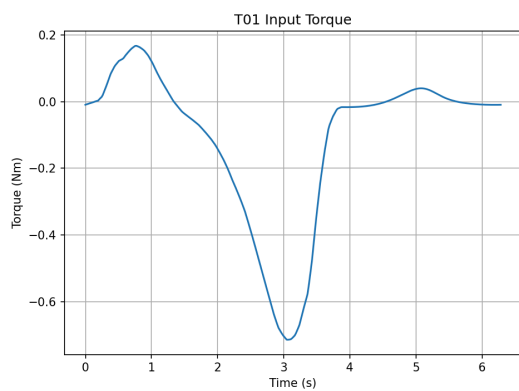
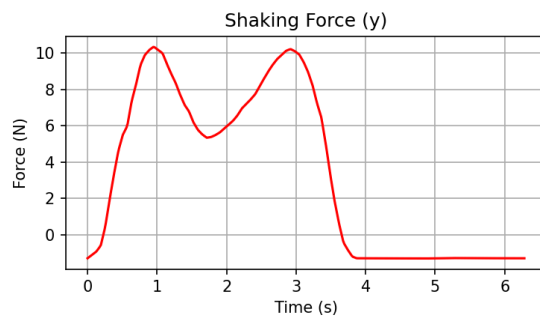
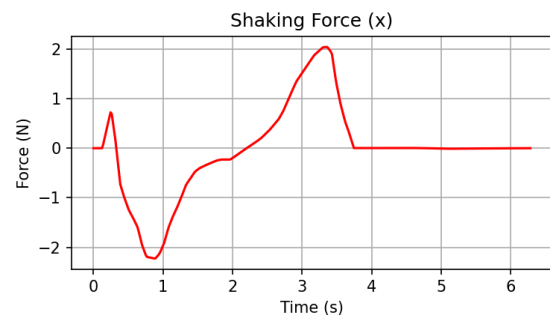
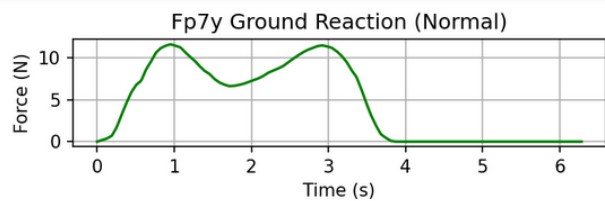
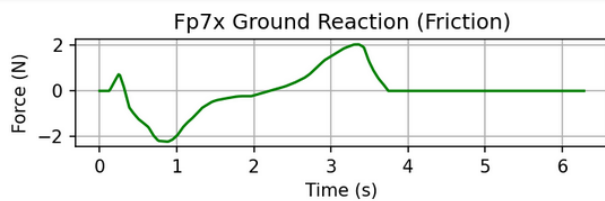
Length: 100mm  
 Height: >120mm



4.  $V_y = 0$  for when foot is in contact with ground:







### Observations:

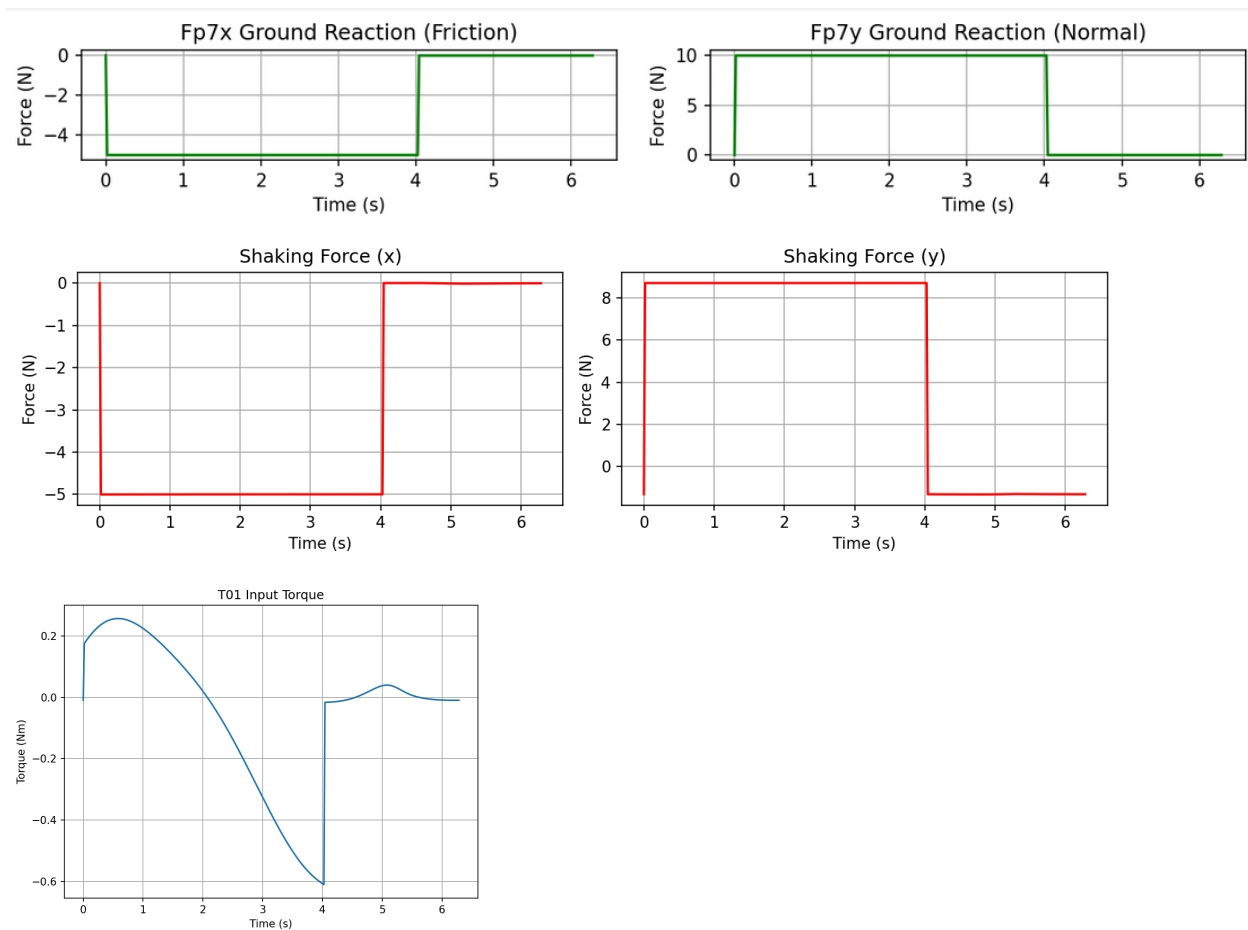
- The shaking forces are periodic, and vary within permissible limits of deformation:

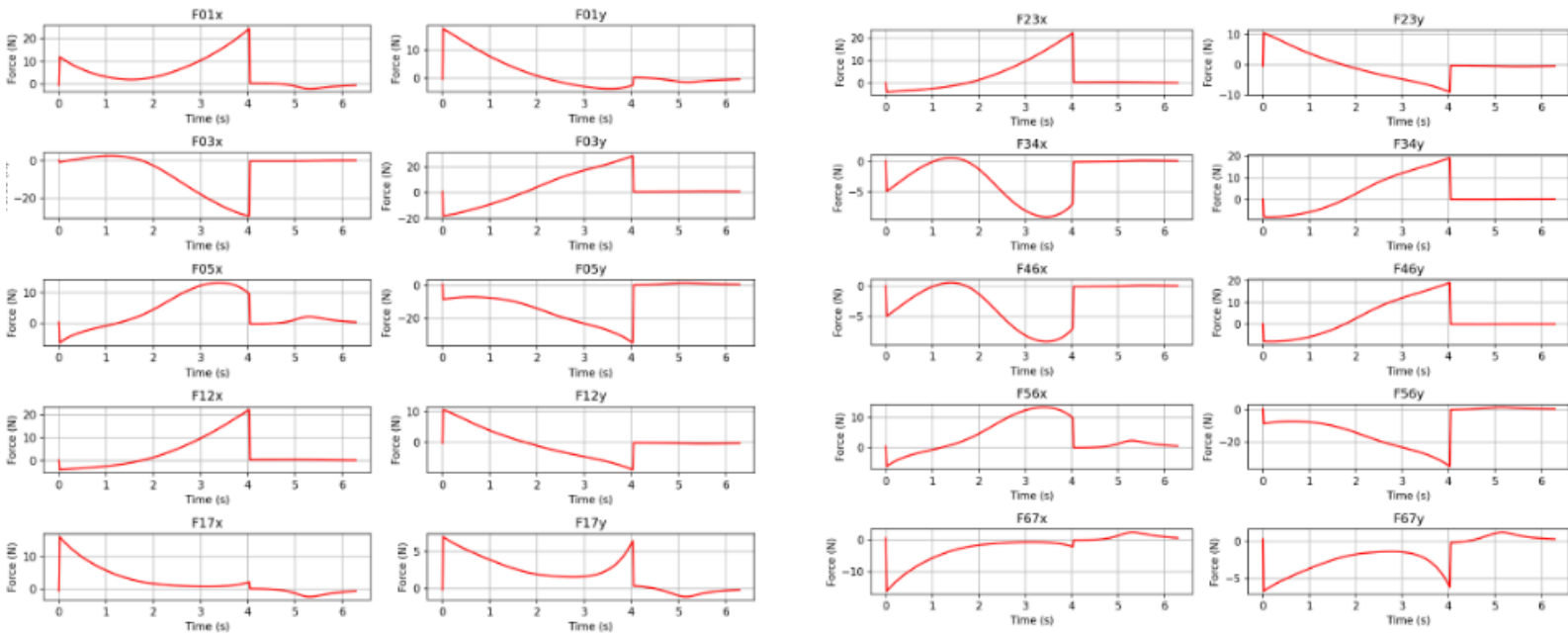
| Shaking Forces | Min(N) | Max(N) |
|----------------|--------|--------|
| x              | -2.25  | 2      |
| y              | -1.8   | 10     |

- The motor's torque characteristics must be capable of handling changes in torque as shown in the plot.

|              | Min(Nmm) | Max(Nmm) |
|--------------|----------|----------|
| Input Torque | -0.75    | 0.15     |

### Foot Subject to Step Input:





#### Observations:

- The shaking forces are periodic, and vary within permissible limits of deformation:

| Shaking Forces | Min(N) | Max(N) |
|----------------|--------|--------|
| x              | -5     | 0      |
| y              | 0      | 10     |

- The motor's torque characteristics must be capable of handling changes in torque as shown in the plot:

|              | Min(Nmm) | Max(Nmm) |
|--------------|----------|----------|
| Input Torque | -0.6     | 0.25     |

#### Simulation:

We are using ADAMS View to verify our analytical results by matching them with the simulation results. We first simulated to verify the kinematic analysis results. For this, we assumed the links have no mass and no forces are acting on them. We set  $I_{yy}$ ,  $I_{xx} = 0$ . The results matched and therefore, we moved on to dynamic analysis,

Assumptions:

Density of links= 1.37 g/cc (3-D printing)(plastic)

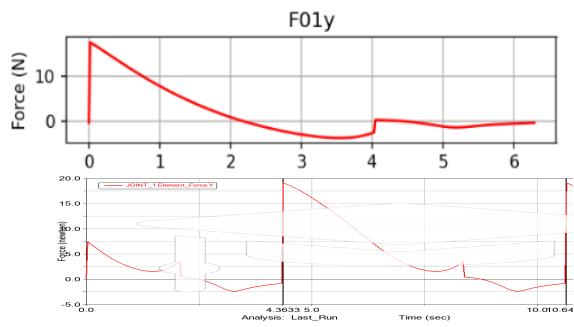
Friction force on the toe = 5 N

Normal force on the toe= 10 N

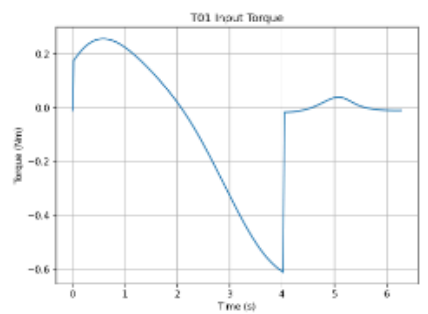
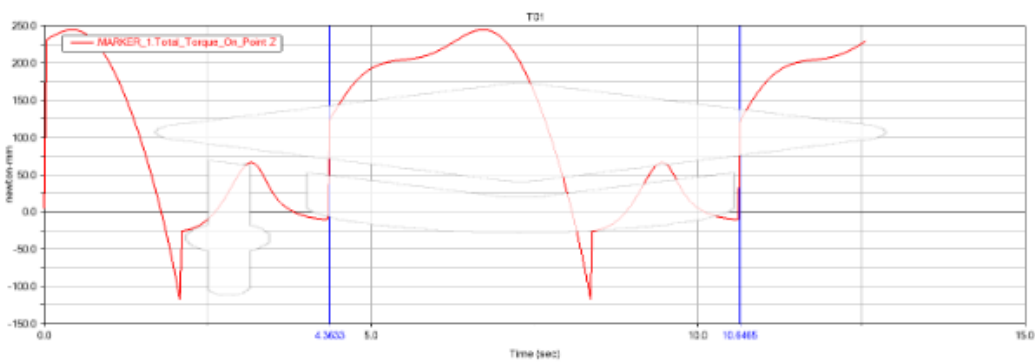
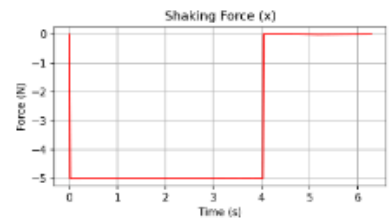
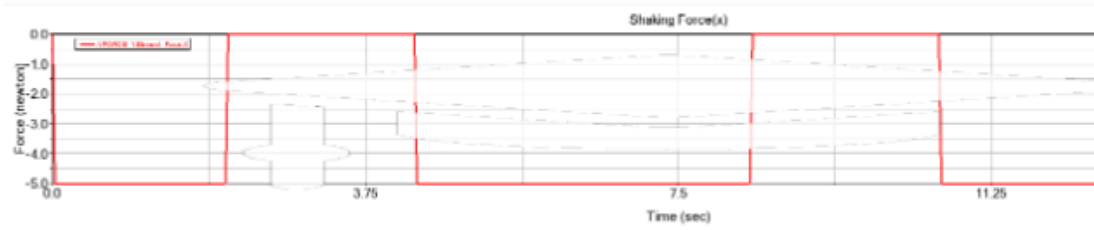
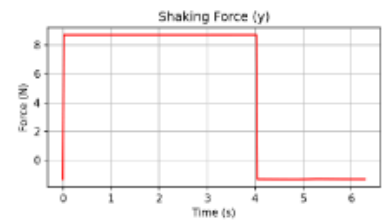
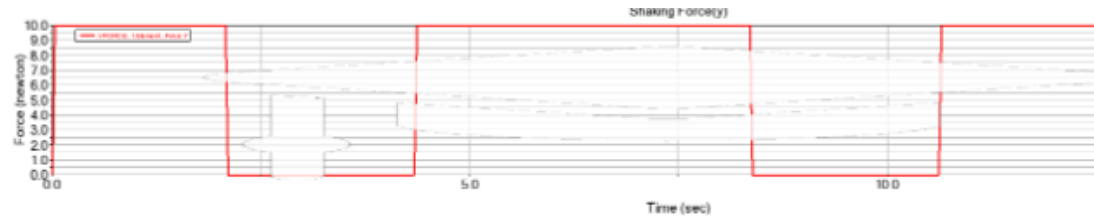
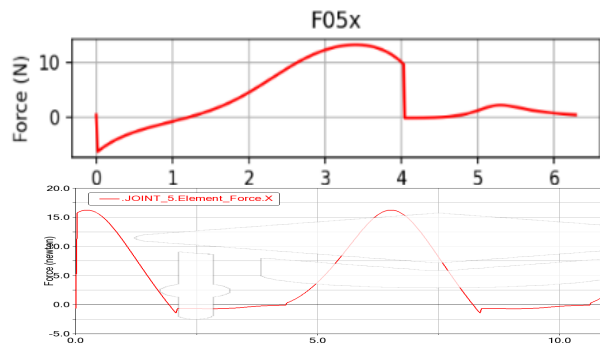
Input angular velocity = 1 rad/s

## Verification:

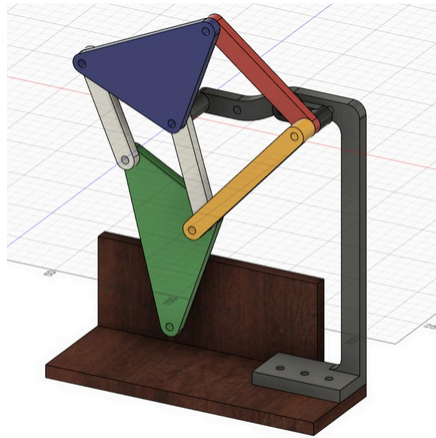
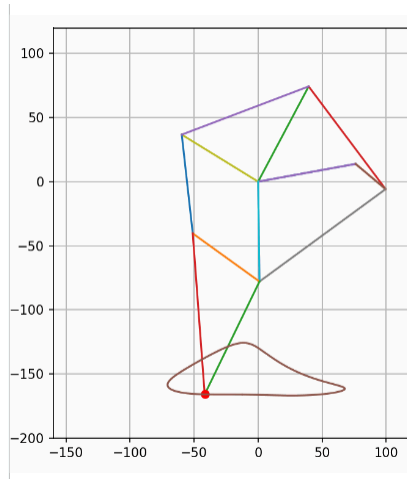
Analytical/ Simulation  $F_{01}$



Analytical / Simulation  $F_{05}$



## Manufacturing and Testing:



### Considerations taken while designing:

- Ensuring no interference between links by pushing interfering links into different planes
- Ensuring weight balance among planes
- Accounting for space taken by bolt heads and nuts, and space to tighten bolts
- Accounting for shrinkage during 3D printing- holes made larger than necessary

## Conclusion:

In this project, we have synthesized, analyzed, simulated and manufactured a functioning prototype for a Jansen Linkage. We have followed an iterative approach to synthesize for 4 precision points (with prescribed timing). The position, velocity and acceleration analyses verified our dimensional synthesis, to meet our requirements (Step Stride and Height, Grashoff Criterion, and Transmission angles). A dynamic analysis was done, providing us with the shaking forces involved and required torque characteristics for a motor at the crank to drive the mechanism on ground.

We have simulated our synthesized linkage using ADAMS View to verify our analytical results and manufactured a functioning prototype of 3D printed links, for demonstration. The model verifies our design constraints.